

When green claims turn brown: The Impact of corporate greenwashing on biodiversity risk in China

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ABSTRACT

Amidst an escalating global biodiversity crisis and a proliferation of corporate environmental claims, this study addresses a critical gap between symbolic corporate communication and its substantive ecological impact. We establish a causal link between corporate greenwashing and increased biodiversity risk using a panel of 17,076 firm-year observations for Chinese listed companies from 2015 to 2024. To ensure robust causal inference, we employ a rigorous identification strategy centered on a Multiple-Difference-in-Differences (M-DID) design, which is complemented by Propensity Score Matching (PSM), an Instrumental Variable (IV) approach, and placebo tests. Our findings reveal that greenwashing causally increases biodiversity risk through a dual-pathway mechanism. Internally, it fosters a "green technology innovation bubble," prioritizing patent quantity over quality. Externally, it degrades "environmental disclosure quality," shielding harmful operations from scrutiny. This detrimental relationship is attenuated by CEOs with green experience but exacerbated by managerial myopia. Our research bridges a crucial theoretical divide between impression management and real-world ecological outcomes, offering vital insights for regulators and investors seeking to curb specious environmentalism.

1. Introduction

A profound dissonance lies at the heart of contemporary corporate sustainability. The natural world is experiencing a catastrophic and accelerated loss of biodiversity, eroding the very ecological foundations of the global economy (Ceballos et al., 2015; Dasgupta, 2021). This has precipitated a concurrent crisis of trust in the environmental pledges of the firms often implicated in this decline. In response to intensifying regulatory and public scrutiny (Boubaker et al., 2024; Hu et al., 2022), corporations have unleashed a surge in environmental disclosures. Yet, this proliferation of green rhetoric has failed to bend the curve of ecological degradation. This schism between corporate communication and biophysical reality motivates a critical inquiry that moves beyond intent to material impact. While the practice of "greenwashing"—conveying misleading impressions about environmental performance—is a well-documented corporate strategy (Laufer, 2003), we lack a clear understanding of its ultimate ecological consequences. Does this symbolic act represent mere reputational posturing, or does it function as a strategic enabler of tangible environmental harm?

The existing literature offers a fractured perspective, having evolved along two parallel but rarely intersecting paths. One extensive stream of

research investigates corporate greenwashing as a tool of impression management, meticulously documenting its antecedents and its consequences for firm reputation and market value (Delmas & Burbano, 2011; Robertson et al., 2023). A separate, rapidly expanding body of work has established that biodiversity risk is not a peripheral ecological issue but a financially material concern for firms (Coqueret et al., 2025; Duong et al., 2025). Recent evidence demonstrates that exposure to biodiversity risk can directly impair firm performance (Bach et al., 2025), diminish operational efficiency (Li et al., 2025), elevate stock price crash risk (Liang et al., 2024), influence corporate cash holdings (Ahmad & Karpuz, 2024), and increase the cost of debt (Li et al., 2025). A critical theoretical gap, however, separates these two conversations. While we increasingly understand the financial consequences of biodiversity risk, we know surprisingly little about its corporate antecedents. This leads us to our central research question: **Does corporate greenwashing causally increase biodiversity risk, and if so, through what mechanisms and under what conditions?**

To answer this question, we construct a unique and powerful research laboratory using the context of China from 2015 to 2024. This setting is ideal for several reasons. First, as the world's second-largest economy and a mega-biodiverse nation facing acute ecological

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pressures, China presents a natural experiment where the incentives for firms to engage in greenwashing are exceptionally high. Second, the country's dynamic, top-down regulatory landscape—marked by major policy shocks like the 2018 creation of the Ministry of Ecology and Environment and the subsequent rollout of mandatory ESG disclosures—provides a compelling quasi-natural experimental design for rigorous causal identification. Finally, by examining these dynamics in a leading emerging market, our study generates critical insights that promise to extend and challenge theories developed primarily within Western institutional frameworks.

This study makes three significant and interlocking contributions. **First**, where prior work has focused on the financial consequences of biodiversity risk (Bach et al., 2025; Liang et al., 2024), we identify a key communicative antecedent. By establishing a causal link from greenwashing to biodiversity risk, we provide large-scale evidence that a firm's symbolic strategy is not decoupled from its real-world impact but instead directly shapes its tangible and financially material ecological footprint. **Second**, we unpack the "black box" of this detrimental relationship by theorizing and testing a novel mediating mechanism: the "green technology innovation bubble." We demonstrate how the strategic imperative to appear green perversely incentivizes superficial, quantity-based innovation over substantive, quality-based solutions. **Third**, our analysis of CEO green experience and managerial myopia as critical boundary conditions refines corporate governance theory. We show how executive oversight and orientation are pivotal in either mitigating or exacerbating the translation of environmental rhetoric into ecological reality. To substantiate these claims, we employ a rigorous identification strategy—combining a Multiple-Difference-in-Differences (M-DID) design with Propensity Score Matching (PSM) and an Instrumental Variable (IV) approach to ensure robust causal inference. Our findings offer vital insights for regulators, investors, and managers seeking to curb symbolic environmentalism and foster genuine corporate accountability.

The remainder of this paper is structured as follows: 2 reviews relevant literature and develops our hypotheses, 3 describes our research design, 4 presents empirical results, and 5 concludes.

2. Literature review and hypothesis development

2.1. Corporate greenwashing and biodiversity risk

An extensive body of work has established corporate greenwashing—the act of conveying misleading impressions about a company's environmental performance (Santos et al., 2024; Torelli et al., 2020)—as a key strategy for managing stakeholder perceptions. The literature provides a clear consensus on its tactics and motivations. Foundational research identified core greenwashing tactics such as confusion, fronting, and posturing (Laufer, 2003), with subsequent work demonstrating how firms strategically use the selective disclosure of positive information to obscure negative environmental impacts, thereby allowing harmful operations to persist (Marquis et al., 2016). The motivations are largely understood to be economic and reputational; firms engage in greenwashing to manage their public image and mitigate reputational threats while avoiding the significant costs associated with substantive environmental improvements (Kim & Lyon, 2015; He et al., 2022). These strategies are executed through sophisticated communication, including vague claims, false certifications (De Freitas Netto et al., 2020), and selective data to project a level of environmental stewardship that exceeds actual performance (Lyon & Montgomery, 2015). Critically, this research stream has primarily framed greenwashing as a symbolic act whose consequences are reputational damage and the erosion of trust, rather than direct ecological harm.

Concurrently, a distinct but equally robust stream of research has established the severe impact of corporate activities on global biodiversity and its financial materiality. Biodiversity risk is defined as the

degree of ecosystem degradation driven by corporate operations, including land use changes, nitrogen deposition, and the release of toxic compounds (Addison et al., 2020; Giglio et al., 2023). Seminal work in this area moved beyond simple attribution, quantifying how international trade and complex corporate supply chains are primary vectors for biodiversity threats (Lenzen et al., 2012; Lambert & Thorlakson, 2018). Recent studies have underscored that this ecological risk translates directly into financial risk, affecting firm performance (Bach et al., 2025; Li et al., 2025), stock price crash risk (Liang et al., 2024), corporate cash holdings (Ahmad & Karpuz, 2024), the cost of debt (Li et al., 2025), and even dividend payouts (Zhou et al., 2025).

Despite the clear financial incentives to manage this risk, a significant gap persists between firms' stated biodiversity commitments and their implemented actions (Boiral et al., 2018). Corporate biodiversity reporting often lacks the specificity and measurable targets required for accountability (Haque & Jones, 2020). This issue is compounded by the inherent difficulty in linking biodiversity-related financial disclosures to real-world conservation outcomes. As de Grefte and de Bruin (2025) argue, the uncertainty surrounding the actual impact of biodiversity finance makes corporate disclosures susceptible to exaggeration and greenwashing. This creates a paradoxical situation where symbolic action can overshadow substantive harm. Strikingly, recent evidence suggests that firms with higher biodiversity risk may exhibit better ESG performance (Pi et al., 2025), indicating that at-risk firms may engage in sophisticated impression management to mask their underlying ecological liabilities.

This brings the two research domains into direct tension. The greenwashing literature focuses on the symbolic act and its reputational fallout, while the biodiversity literature focuses on the operational act and its ecological and financial consequences. What remains a "black box" is the causal pathway from the symbolic act of greenwashing to the tangible outcome of biodiversity degradation. This leaves a crucial, unanswered question: Is greenwashing merely "cheap talk" with reputational consequences, or is it a strategic enabler that actively facilitates and exacerbates substantive ecological harm, potentially explaining the paradoxes observed in corporate reporting?

We argue the latter. The theoretical logic connecting them operates through several pathways. First, greenwashing creates a facade of environmental progress that diverts stakeholder attention from a firm's actual ecological impacts, thereby reducing external pressure for meaningful conservation (Wang et al., 2024). Second, resources allocated to elaborate greenwashing campaigns represent a significant opportunity cost, diverting funds that could have been invested in substantive biodiversity protection (Pizzetti et al., 2021). Third, greenwashing institutionalizes "means-ends decoupling," where symbolic environmental initiatives become organizationally disconnected from the core business operations that degrade ecosystems (Bromley & Powell, 2012).

Early evidence supports this connection, showing that firms engaging in symbolic environmental management often continue environmentally destructive operations (Boiral & Heras-Saizarbitoria, 2017). Successful impression management can allow companies to delay or circumvent operational constraints that would otherwise protect biodiversity (Nyilasy et al., 2014). For example, firms with prominent environmental marketing have been found to operate in high-biodiversity regions while implementing minimal protective measures (Hassan et al., 2020). By creating a misleading narrative, greenwashing provides an "information shield" that allows tangible harm to continue and expand. To formally test this causal link, we propose our first hypothesis:

H1. : Corporate greenwashing is positively associated with increased biodiversity risk.

2.2. The moderating role of CEO green experience and managerial myopia

The relationship between a firm's symbolic environmental strategy and its substantive ecological impact is not uniform; it is shaped by the internal governance context in which decisions are made. While general governance mechanisms, such as board independence, offer a layer of oversight, we argue that the translation of greenwashing into tangible biodiversity risk is particularly sensitive to the characteristics of the chief executive. The strategic choice to pursue symbolic communication over substantive investment involves complex, technically nuanced, and often intertemporal trade-offs that are ultimately filtered through the CEO's cognitive and motivational lens.

Therefore, we move beyond broad governance metrics to focus on two core dimensions of executive orientation that profoundly shape how these trade-offs are resolved: the CEO's specialized capability and values (proxied by green experience) and their temporal orientation (proxied by managerial myopia). Together, these constructs capture the key decision-maker's disposition and capacity to align a firm's green rhetoric with its operational reality.

CEO green experience refers to a chief executive's prior involvement in environmental initiatives, sustainability-focused education, or leadership roles in environmentally responsible organizations (Li et al., 2024; Huang & Wei, 2023). In line with upper echelons theory, this experiential background is instrumental in shaping an executive's cognitive base, values, and decision-making when confronting complex ecological challenges (Quan et al., 2021). A CEO with authentic green experience brings specialized environmental knowledge that is valuable in two respects.

First, it enhances their cognitive capability to distinguish between substantive environmental actions and purely symbolic greenwashing (Lin et al., 2025). Biodiversity issues are technically complex and their financial implications are often long-term and uncertain, making them opaque to non-expert leaders. An experienced CEO can more accurately assess the ecological risks of operational decisions and recognize when a proposed "green" initiative lacks substantive merit. Second, this background often correlates with stronger intrinsic environmental values, motivating a commitment to genuine ecological protection that transcends simple reputational management.

Prior research lends strong support to this view. Zhang et al. (2024) found that executives with environmental experience prioritized actual environmental performance improvements over symbolic communications. Similarly, Haojun and Jiazhu (2024) documented that environmentally knowledgeable CEOs allocated greater resources to verified biodiversity conservation measures rather than to superficial marketing claims. This suggests that green experience can function as an important internal control mechanism. When an environmentally experienced executive leads an organization, they are more likely to challenge the decoupling of symbolic claims from operational reality. Even if the firm engages in some level of green communication, such a leader is better equipped and more motivated to ensure that these initiatives are tethered to genuine actions that mitigate, rather than exacerbate, biodiversity risk. Therefore, we hypothesize:

H2a. CEO green experience negatively moderates the relationship between corporate greenwashing and biodiversity risk, such that the positive relationship is weaker when CEO green experience is high.

Managerial myopia represents a short-term orientation in executive decision-making that prioritizes immediate financial outcomes over long-term organizational sustainability (Laverty, 1996). This creates a critical contingency for the ecological consequences of greenwashing, as biodiversity protection inherently involves a temporal trade-off: incurring immediate, certain costs for long-term, uncertain environmental benefits. Greenwashing allows managers to exploit this trade-off by securing short-term reputational gains without making the

corresponding long-term investments (Bansal & DesJardine, 2014).

We argue that managerial myopia acts as a powerful cognitive and incentive-based mechanism that amplifies this detrimental trade-off. A myopic management team, driven by pressure for quarterly earnings and immediate performance metrics, is predisposed to heavily discount future ecological risks (Slawinski & Bansal, 2015). For these managers, greenwashing is not just a communication tactic but an optimal short-term solution. Once the symbolic act of greenwashing has been performed and the immediate reputational benefits are secured, a myopic manager is significantly less likely to approve the costly, long-term operational changes required for genuine biodiversity protection, as these actions offer no immediate, tangible rewards. The "green" image has been achieved, and any further investment is seen as an unnecessary cost that detracts from short-term profits.

In contrast, a non-myopic manager, while still potentially engaging in greenwashing, may retain a strategic view of long-term risk and feel compelled to allocate at least some resources to substantive environmental actions to hedge against future regulatory or physical risks. Managerial myopia effectively removes this residual long-term strategic thinking. It intensifies the decoupling between a firm's green rhetoric and its operational reality, thereby strengthening the positive association between greenwashing and actual biodiversity degradation. Therefore, we hypothesize:

H2b. Managerial myopia positively moderates the relationship between corporate greenwashing and biodiversity risk, such that the positive relationship is stronger when managerial myopia is high.

2.3. The mediating role of green technology innovation bubble and environmental disclosure quality

To unpack the "black box" linking the symbolic act of greenwashing to substantive ecological harm, we theorize that the effect operates through two primary and distinct pathways: an internal operational pathway and an external informational pathway. These channels represent the principal ways in which a firm's communicative strategy translates into real-world consequences. The internal pathway concerns the misallocation of organizational resources, while the external pathway involves the manipulation of the firm's information environment.

The first mechanism, the green technology innovation bubble, captures how greenwashing distorts a firm's internal resource allocation and innovation strategy. We define this "bubble" as a phenomenon wherein firms, driven by impression management, prioritize the quantity of environmental innovation initiatives over their substantive quality or ecological effectiveness (Geng et al., 2024). This creates a facade of environmental commitment that is not backed by genuine ecological benefits.

This concept builds on foundational distinctions between genuine eco-innovations that confer competitive and environmental benefits (Porter & van der Linde, 1995) and symbolic technologies deployed primarily for optics (Aragón-Correa & Rubio-Lopez, 2007). Corporate greenwashing provides fertile ground for such a bubble to form. When a firm's primary goal is to project a green image rather than achieve environmental impact, its innovation efforts are naturally channeled toward projects that are visible, easily quantifiable, and generate positive announcements, rather than those that require deep, disruptive operational changes (Yuan et al., 2023; Ziegler et al., 2022). As documented by Bansal and Clelland (2004), firms under legitimacy pressure often respond with a scattered portfolio of minor innovations that signals commitment without addressing their core ecological externalities.

This quantity-over-quality approach to innovation contributes directly to biodiversity risk. By spreading limited R&D resources across numerous superficial projects, firms fail to achieve the depth of investment needed for meaningful ecological protection (Delmas & Toffel,

2008). Moreover, these innovation bubbles often form around technologies that do not disrupt the firm's core business model—precisely the operations most responsible for its biodiversity impacts. Recent work by [Tian and Chen \(2025\)](#) supports this, finding that firms pursuing multiple but superficial eco-innovations performed worse on biodiversity metrics than firms implementing fewer but more substantive ecological projects. This suggests that greenwashing does not just coexist with ecological harm; it actively facilitates it by misdirecting the very resources that could be used for its mitigation. We therefore propose:

H3a. Green technology innovation bubble mediates the relationship between corporate greenwashing and biodiversity risk, such that corporate greenwashing increases green technology innovation bubble, which in turn increases biodiversity risk.

The second mechanism operates through the external information environment. Environmental disclosure quality—representing the comprehensiveness, accuracy, and relevance of environmental information provided to stakeholders (Clarkson et al., 2008)—is not merely a consequence of greenwashing, but is constitutive of the act itself. Greenwashing is fundamentally an exercise in managing perceptions by degrading the quality of information available to outside observers.

The evolution of disclosure research highlights this strategic dimension, moving from early content analyses (Wiseman, 1982) to legitimacy-based theories of reporting (Deegan & Rankin, 1996) and demonstrations of the disconnect between disclosure and performance (Patten, 2002). Firms engaging in greenwashing employ sophisticated tactics to lower disclosure quality, such as "thematic manipulation"—emphasizing minor positive actions while obscuring major negative impacts (Bowen, 2014)—and "calculated ambiguity" to prevent meaningful assessment (Milne et al., 2009).

This degradation of disclosure quality serves as a direct channel to increased biodiversity risk for several reasons. First, misleading disclosures create information asymmetries that impair the ability of stakeholders—investors, regulators, and communities—to accurately assess a firm’s ecological footprint and exert pressure for change (Enríquez-de-Salamanca, 2018). Second, low-quality information can distort capital allocation, channeling investment toward firms with significant but unrecognized biodiversity liabilities (Ozkan et al., 2023). Third, a pattern of superficial reporting reinforces an internal culture that prioritizes impression management over substantive ecological stewardship (Nor Ahmad et al., 2022).

Recent research provides direct empirical support for this pathway. In a landmark study, [Marquis and Qian \(2014\)](#) found that Chinese firms used symbolic philanthropic disclosures, such as donating to

government-favored causes, to build political legitimacy. This legitimacy subsequently allowed them to operate with greater impunity, leading to more environmental violations. This logic also applies to environmental reporting, as other studies suggest that firms with elaborate but low-quality disclosures are able to expand into sensitive habitats with less stakeholder resistance (Qin et al., 2024). Together, these findings illustrate how various symbolic actions create an "information shield," a mechanism that disables the market and social pressures that would otherwise constrain environmentally harmful behavior. By creating a misleading narrative, greenwashing provides this shield, allowing tangible harm to continue and expand. To formally test this causal link, we propose our first hypothesis:

H3b. Environmental disclosure quality mediates the relationship between corporate greenwashing and biodiversity risk, such that corporate greenwashing leads to lower environmental disclosure quality, which in turn increases biodiversity risk.

The specific research framework of this study is shown in Fig. 1.

3. Research design

3.1. Data and sample

Our empirical analysis employs a comprehensive panel dataset of Chinese listed companies spanning from 2015 to 2024. We select this timeframe for three key reasons: (1) it begins with China's implementation of the "Ecological Civilization" national strategy in 2015, marking a pivotal shift in environmental governance (Jiang et al., 2020; Wu et al., 2021); (2) it encompasses critical regulatory reforms including the 2018 Ministry of Ecology and Environment restructuring and 2020 mandatory ESG disclosure requirements (Wen, 2020); and (3) it provides sufficient longitudinal data to capture both short-term fluctuations and long-term trends in greenwashing practices and biodiversity impacts.

We source financial data from the CSMAR database, environmental information from CNRDS, and greenwashing metrics from manually coded corporate reports. Our primary data collection covers the period from 2015 to 2023, for which complete annual reports were available. To extend the time series to 2024 and ensure a consistent panel structure for our analysis, we estimated the data for the final year. Specifically, we employed a multivariate imputation by chained equations (MICE) approach. This method generates robust forecasts by leveraging the variable interdependencies and time-series patterns observed in the historical 2015–2023 data, thereby preserving the underlying data

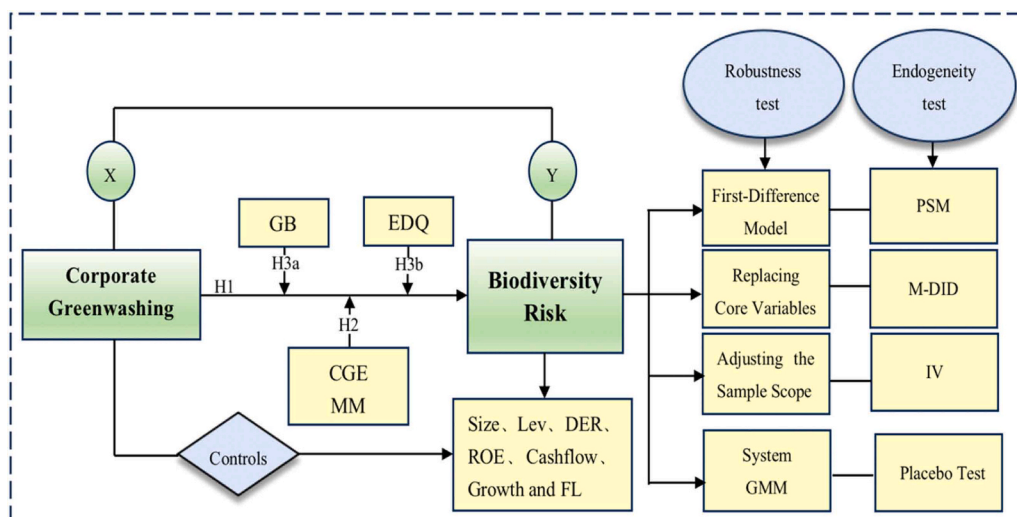


Fig. 1. Overall research framework

structure while extending the panel.

Following established practice in Chinese capital market research (Liu et al., 2019; Titman et al., 2022), we apply several filtering criteria: (1) excluding ST and *ST firms (Special Treatment designated companies with financial distress or regulatory violations) to remove firms facing significant survival concerns that could distort environmental priorities; (2) eliminating financial sector firms due to their distinct regulatory environment and business models; and (3) applying Winsorization at the 1st and 99th percentiles for continuous variables to mitigate the influence of outliers while preserving sample variability. The final balanced panel dataset comprises 17,076 firm-year observations across 1898 unique firms.

3.2. Variable definition

3.2.1. Independent variable

Building on signaling theory and impression-management insights, this study proposes a rigorously specified, multidimensional metric of corporate greenwashing. We frame greenwashing as the deliberate misrepresentation of a firm's environmental performance across four organisational domains—governance, operations, resource allocation, and regulatory compliance (Zhang, 2023). The construct integrates two orthogonal components: selective-disclosure intensity (GWLS) and expressive-manipulation magnitude (GWLE).

To capture the extent to which firms withhold material information, we compute a materiality-weighted omission rate:

$$GWLS = 100 \times \frac{\sum_i w_i N_i}{\sum_i w_i} \quad (1)$$

Where, w_i = expert-validated materiality weight for disclosure item i (1–5 scale), N_i = 1 if item i is omitted, 0 otherwise

Weighting omissions by their materiality penalises concealment of the high-priority items most valued by stakeholders (Galletta et al., 2024).

Using a human-machine hybrid text-analytics pipeline (Krippendorff's $\alpha \geq 0.85$), we classify every sentence in the environmental report as symbolic (α), partially substantive (β), or fully substantive (γ). The resulting score is

$$GWLE = 100 \times \frac{\sum (2\alpha + \beta)}{2 \times \sum (\alpha + \beta + \gamma)} \quad (2)$$

Symbolic statements—typically vague, unverifiable commitments—receive double weight to reflect their greater capacity to mislead.

To obtain a single, cross-sector-comparable indicator, we combine GWLS and GWLE via a modified geometric mean:

$$GW = (GWLS^k \times GWLE^k)^{1/k} \quad (3)$$

3.2.2. Dependent variable

Biodiversity Risk (BR). This study operationalizes corporate biodiversity risk exposure as a binary variable (BR) that categorizes firms based on their potential for ecological impact (Li et al., 2025; Giglio et al., 2023). Acknowledging the intractability of directly measuring the precise, firm-specific ecological impact across a large-scale panel, we construct a robust proxy for a firm's exposure to such risks. To achieve this, we follow established methodologies in sustainability accounting by leveraging the Sustainability Accounting Standards Board (SASB) Materiality Map (Tian & Chen, 2025). The theoretical underpinning of this approach is that firms operating in industries where ecological impacts are deemed financially material face a systematically higher level of biodiversity risk exposure, as their fundamental business models are intrinsically linked to activities such as significant land use, resource extraction, or high-impact emissions. This method provides an objective

and structural assessment of risk that is grounded in an investor-focused, globally recognized standard, thereby mitigating the measurement biases inherent in relying on firm self-disclosures, which can be subject to impression management. The variable is constructed as follows:

$$BR_{i,t} = \begin{cases} 1, & \text{if } \sum_{j=1}^n \omega_j \cdot (SASB_{i,j} \times EI_{i,j,t}) > \tau \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Here, $BR_{i,t}$ represents the biodiversity risk exposure for firm i in year t . $SASB_{i,j}$ indicates whether the firm's subsector is identified by SASB as having material disclosure topics under the "ecological impacts" category. For implementation, industries meeting this criterion were coded as high-risk (1), while all others were coded as low-risk (0). To ensure comprehensive and accurate classification, these SASB industry definitions were meticulously mapped to the Bloomberg Industry Classification System. This approach ensures our measurement is not only objective but also captures the structural risk embedded in a firm's core operations, providing a consistent and valid proxy for the dependent variable.

3.2.3. Moderating variable

(1) CEO green experience

CEO green experience (CGE) indicates whether the chief executive officer has extensive professional exposure to or substantial expertise in environmental sustainability issues (Wang et al., 2024). It is operationalized as a binary variable, where a value of 1 denotes environmental significant knowledge, and 0 represents its absence. To ensure accuracy and objectivity, we code CEOs as possessing relevant green experience if their professional history demonstrates leadership roles in sustainability initiatives, formal education credentials in environmental science or sustainability-focused business disciplines, board memberships within recognized environmental organizations, significant contributions to environmental policy discussions, or explicit managerial responsibility for major corporate sustainability transformations. This binary classification effectively differentiates CEOs according to their demonstrated engagement and professional credentials in sustainability matters.

(2) Moderating Variables

Managerial myopia (MM) characterizes executives' tendency to prioritize immediate performance outcomes rather than investing strategically in long-term sustainable growth (Liu & Zhang, 2023). Building upon prior textual analytical methods (Ding et al., 2024), this study quantifies managerial myopia by analyzing the language patterns in Management Discussion and Analysis (MD&A) sections of firms' annual reports. Specifically, we deploy a Word2Vec computational text analysis model based on the Continuous Bag-of-Words (CBOW) algorithm to identify linguistic markers indicating a short-term strategic orientation. Common temporal terms and phrases associated with short-termism—such as "immediate," "as soon as possible," "months," "days," "urgent," and "pressure"—are systematically extracted and analyzed. The managerial myopia score is expressed as the percentage of short-term oriented terms relative to the total word count in the MD&A section, thus providing a robust, textual proxy for executives' short-term orientation.

$$Myopia_{i,t} = \left(\frac{\text{Short-term Words Count}_{i,t}}{\text{Total MD\&A Words Count}_{i,t}} \right) \times 100 \quad (5)$$

3.2.4. Mediating variable

(1) Green technology innovation bubble

Building on advanced methodologies in innovation assessment,

this study measures Green Technology Innovation Bubbles (GB) through a multidimensional framework that captures the gap between innovation quantity and substantive quality. While green patents indicate innovation output (Geng et al., 2024), conventional metrics often overlook the distinction between superficial patenting and genuine advancement.

- a. The Patent Quantity-Quality Differential Index (PQDI), which measures standardized disparities between patent applications and authorizations with a temporal decay factor (λ):

$$PQDI_{it} = \frac{\sum_{\tau=1}^T (PA_{it} - PG_{it})e^{-\lambda(t-\tau)}}{\sigma_{industry} \sqrt{\sum_{\tau=1}^T e^{-2\lambda(t-\tau)}}} \quad (6)$$

- b. The Patent Citation-Weighted Index (PCWI), incorporating citation metrics:

$$PCWI_{it} = \frac{\sum_{k=1}^{PG_{it}} \left(\frac{C_{ik}}{E[C_{industry,k}]} + \sqrt{F_{ik}} \right)^\alpha}{PG_{it}^{1-\alpha}} \quad (7)$$

- c. The Innovation Efficiency Index (IEI), contextualizing outputs against resource inputs:

$$IEI_{it} = \left(\frac{\sum_{\tau=1}^T PG_{it}e^{-\lambda(t-\tau)}}{K_{it}^\theta H_{it}^{1-\theta}} \right) \left(\frac{Y_{it}}{Y_{industry}} \right)^\beta \quad (8)$$

These are synthesized into a composite GB measure:

$$GB_{it} = \left[\frac{1}{3} \sum_{j=1}^3 \left(\frac{I_{jit}}{\mu_j} \right)^\varphi \right]^{1/\varphi} \cdot \exp \left(\frac{Skewness(I_{jit})}{3} \right) \quad (9)$$

(2) Environmental disclosure quality

We categorized environmental disclosures based on monetization characteristics to assess disclosure quality comprehensively. For monetized information (Environmental Liability Disclosure and Environmental Governance categories), disclosures combining quantitative and qualitative elements received 2 points, purely qualitative disclosures 1 point, and no disclosure 0 points (Qian & Chen, 2021). For non-monetized information (Environmental Management, Certification, and Disclosure Medium categories), disclosed indicators received 2 points while undisclosed indicators received 0 points. Our framework encompasses five dimensions with 25 distinct scoring items. After evaluation, we summed individual scores and applied logarithmic transformation to derive our Environmental Disclosure Quality (EDQ) metric. This transformation addresses potential distribution skewness and emphasizes relative differences in disclosure practices across firms.

The resulting EDQ index holistically reflects corporate environmental transparency, capturing both comprehensiveness and information quality. Detailed evaluation criteria and scoring methodology are presented in Appendix B, Table B1.

3.2.5. Control variable

Following established literature (Balasubramanian et al., 2021; Khan, 2022), we control for several firm characteristics that might influence biodiversity risk. Firm size (Size) is measured as the natural logarithm of total assets. Financial leverage (Lev) is calculated as total liabilities divided by total assets. Debt-to-equity ratio (DER) represents financial risk exposure. Return on equity (ROE) captures profitability.

Operating cash flow (Cashflow) controls for financial flexibility. Sales growth (Growth) accounts for expansion trajectories that might impact environmental priorities. Finally, financial liquidity (FL) measures short-term solvency that could affect environmental investment capacity. The detailed classification and definitions of variables are presented in Table 1.

3.3. Model specification

3.3.1. Benchmark regression model construction

To empirically test our main hypothesis (H1), we employ a two-way fixed effects panel regression model. This specification is chosen to mitigate concerns about omitted variable bias by controlling for two primary sources of unobserved heterogeneity. First, the inclusion of year fixed effects (YEAR_t) accounts for any macroeconomic trends or shocks (e.g., shifts in national environmental policy, business cycles) that are common to all firms in a given year. Second, the industry fixed effects (IND_j) absorb all time-invariant differences across industries, including their baseline ecological sensitivity, technological intensity, and typical regulatory burden. By incorporating these fixed effects, we can more effectively isolate the firm-specific impact of greenwashing from broader temporal and industry-level confounding factors. The baseline model is defined as:

$$BR_{i,t} = \beta_0 + \beta_1 GW_{i,t} + \sum_k \beta_k \text{CONTROLS}_{i,t} + \sum_j \gamma_j \text{IND}_j + \sum_t \delta_t \text{YEAR}_t + \varepsilon_{i,t} \quad (10)$$

Where $BR_{i,t}$ denotes the biodiversity risk exposure for firm i in year t ; $GW_{i,t}$ is the composite greenwashing score; $\text{CONTROLS}_{i,t}$ is a vector of firm-level control variables; IND_j and YEAR_t represent industry and year fixed effects, respectively. A positive and statistically significant β_1 would confirm that greenwashing aggravates biodiversity risk.

3.3.2. Moderating effect regression model construction

To explore the moderating roles of CEO green experience (H2a) and managerial myopia (H2b), we extend the baseline model with interaction terms. The specification is as follows:

$$\begin{aligned} BR_{i,t} &= \beta_0 + \beta_1 GW_{i,t} + \beta_2 CGE_{i,t} + \beta_3 MM_{i,t} + \beta_4 (GW_{i,t} \times CGE_{i,t}) \\ &+ \beta_5 (GW_{i,t} \times MM_{i,t}) + \sum_k \beta_k \text{CONTROLS}_{i,t} + \sum_j \gamma_j \text{IND}_j + \sum_t \delta_t \text{YEAR}_t + \varepsilon_{i,t} \end{aligned} \quad (11)$$

Where $CGE_{i,t}$ captures CEO green experience and $MM_{i,t}$ represents managerial myopia. The interaction terms β_4 and β_5 test the conditional effects of greenwashing under varying managerial characteristics. We expect $\beta_4 < 0$, suggesting that CEO green experience weakens the link between greenwashing and biodiversity risk, while $\beta_5 > 0$ indicates that managerial myopia intensifies this relationship.

3.3.3. Mediating effect regression model construction

To test the mediating mechanisms proposed in H3a and H3b, we adopt the causal steps approach of Lu et al. (2022) and Feng et al. (2024). Specifically, we assess whether greenwashing affects two mediators: the green technology innovation bubble (GB) and environmental disclosure quality (EDQ). The mediation models are expressed as:

$$GB_{i,t} = \alpha_0 + \alpha_1 GW_{i,t} + \sum_k \alpha_k \text{CONTROLS}_{i,t} + \sum_j \gamma_j \text{IND}_j + \sum_t \delta_t \text{YEAR}_t + \varepsilon_{i,t} \quad (12)$$

$$EDQ_{i,t} = \theta_0 + \theta_1 GW_{i,t} + \sum_k \theta_k \text{CONTROLS}_{i,t} + \sum_j \gamma_j \text{IND}_j + \sum_t \delta_t \text{YEAR}_t + \varepsilon_{i,t} \quad (13)$$

Where $GB_{i,t}$ reflects the level of green innovation bubbles and $EDQ_{i,t}$ captures the quality of environmental disclosure.

4. Result

4.1. Descriptive statistics

Table 2 presents descriptive statistics for all variables. Our sample comprises 17,076 firm-year observations from Chinese listed companies between 2015 and 2024. The mean value of Biodiversity Risk (BR) is 0.483, indicating substantial ecological impacts across the sample. Corporate Greenwashing (GW) averages 0.642, suggesting prevalent misleading environmental communications consistent with Marquis et al.'s (2016) findings in emerging markets. The Green Technology Innovation Bubble (GB) mean of 0.018 indicates significant quantity-quality divergence in environmental innovations, aligning with [Berone et al.'s \(2017\)](#) observations. Control variables reveal a moderate leverage ratio (0.457), ROE (0.086), and growth rate (0.173), reflecting a financially stable sample with typical Chinese market characteristics. This pattern of high greenwashing prevalence coupled with substantial biodiversity risk provides preliminary support for our theoretical framework.

4.2. Baseline regression

Table 3 presents the baseline regression results examining the relationship between corporate greenwashing and biodiversity risk. Model 1 includes only the main independent variable with year and industry fixed effects, while Model 2 incorporates all control variables. The results provide strong support for Hypothesis 1, which predicts a positive association between corporate greenwashing and biodiversity risk. In both models, the coefficient of greenwashing (GW) is positive and statistically significant at the 1 % level. Specifically, in the fully specified model (Column 2), the coefficient of GW is 0.317 ($p < 0.01$), indicating a substantial positive effect of greenwashing on biodiversity risk.

This finding aligns with our theoretical framework suggesting that firms engaging in greenwashing are more likely to continue environmentally harmful operations that adversely affect biodiversity. The result is consistent with prior studies by [Zhang et al. \(2024\)](#) and [Hu & Chen \(2025\)](#), who documented how symbolic environmental management allows companies to maintain operations with significant ecological impacts.

4.3. Robustness test

To ensure the robustness and credibility of our baseline findings, we conducted a series of supplementary analyses employing alternative model specifications, variable operationalizations, and sample adjustments. Table 4 reports these robustness tests, with Column (1) presenting the baseline results for reference.

First, we employed a first-difference (FD) model to mitigate potential omitted variable bias and control for time-invariant firm-level heterogeneity. This approach examines year-over-year changes rather than levels, thereby isolating within-firm temporal variations ([Wooldridge, 2010](#)). The FD model is specified as:

$$\Delta BR_{i,t} = \beta_0 + \beta_1 \Delta GW_{i,t} + \sum \beta_k \Delta CONTROLS_{i,t} + \sum \delta_t YEAR_t + \Delta \varepsilon_{i,t} \quad (14)$$

Second, to address potential reverse causality between greenwashing and biodiversity risk, we employed a lagged independent variable specification (Column 3), incorporating a one-year lag of greenwashing. This model assesses whether prior-period greenwashing behaviors influence current biodiversity risk, thus providing a stricter causal inference:

$$BR_{i,t} = \beta_0 + \beta_1 GW_{i,t-1} + \sum \beta_k CONTROLS_{i,t} + \sum \gamma_j IND_j + \sum \delta_t YEAR_t + \varepsilon_{i,t} \quad (15)$$

Third, we tested alternative measurements for our core variables to verify that our results are not contingent on a single operationalization. In Column (4), we adopted an alternative measurement of greenwashing (GW_ALT), drawing on the symbolic-substantive gap framework proposed by [Chen et al. \(2025\)](#). Furthermore, in Column (5), we substituted our dependent variable with an alternative biodiversity risk measure (BR_ALT) based on the textual analysis methodology of [He et al. \(2024\)](#). This method constructs a firm-specific index from annual reports by analyzing the frequency and proportion of biodiversity-related keywords, offering a different lens to quantify a firm's biodiversity risk exposure.

Fourth, to account for potential distortions arising from the COVID-19 pandemic, we re-estimated our model after excluding observations from 2020 and 2021 (Column 6). The pandemic disrupted global operations and altered firms' environmental strategies and disclosure practices ([Wang et al., 2025](#)), which may confound our analysis. Restricting the sample to pre- and post-pandemic periods ensures that our findings are not driven by such anomalies.

Finally, we implemented a two-step system GMM estimator to comprehensively address endogeneity concerns, including potential dynamic panel bias and reverse causality ([Blundell & Bond, 1998](#)). This model uses lagged levels and differences of the endogenous regressors as instruments, controlling for unobserved heterogeneity:

$$BR_{i,t} = \beta_0 + \beta_1 BR_{i,t-1} + \beta_2 GW_{i,t} + \sum \beta_k CONTROLS_{i,t} + \eta_i + \varepsilon_{i,t} \quad (16)$$

Across all six robustness tests, the coefficient on the greenwashing variable remains positive and statistically significant, confirming the stability and reliability of our main findings. In the FD model (Column 2), the estimated coefficient (0.097) is slightly attenuated, as expected, but still significant. The lagged model (Column 3) yields a larger coefficient (0.187), suggesting that the adverse effects of greenwashing may intensify over time. The results using an alternative greenwashing measure (0.134 in Column 4) and an alternative biodiversity risk measure (0.145 in Column 5) remain highly significant, demonstrating robustness to different variable operationalizations. Excluding pandemic-affected years (Column 6) yields the strongest coefficient (0.203), implying that the relationship is even more pronounced under typical business conditions. The system GMM estimator (Column 7) produces a coefficient of 0.116, reinforcing our conclusions after controlling for complex endogeneity issues.

4.4. Endogeneity test

4.4.1. Propensity score matching

To address potential selection bias, we implemented propensity score matching to compare firms with similar characteristics but different greenwashing levels. We classified firms with greenwashing scores above the median as the treatment group and those below as the control group.

Using nearest-neighbor matching with caliper (0.02) and no replacement, we achieved good covariate balance (mean standardized bias reduced from 23.7 % to 4.8 %). Table 5 presents regression results using the matched sample of 6328 pairs. The coefficient for high greenwashing remains positive and significant (0.133 without controls, 0.117 with controls), though slightly attenuated compared to our main results.

This suggests that while selection effects partially explain the relationship, greenwashing has a distinct impact on biodiversity risk beyond correlation with firm characteristics. These findings strengthen our causal inference by demonstrating robustness to potential selection bias.

4.4.2. Multiple-difference-in-differences (M-DID)

To further substantiate a causal link between greenwashing and biodiversity risk, we employ a multiple-difference-in-differences (M-DID) research design. This approach leverages two significant, staggered regulatory reforms in China as quasi-natural experiments: (1) the 2018 institutional reform establishing the Ministry of Ecology and Environment (MEE), which centralized environmental governance (Lin, 2021); and (2) the 2020 mandatory ESG disclosure requirements for listed firms (Krueger et al., 2024). These exogenous shocks allow us to causally evaluate how shifts in the regulatory landscape alter the consequences of corporate greenwashing. Our M-DID model is specified as follows:

$$BR_{i,t} = \beta_0 + \beta_1 GW_{i,t} + \beta_2 Post_{MEE,t} + \beta_3 Post_{ESG,t} + \beta_4 (GW_{i,t} \times Post_{MEE,t}) + \beta_5 (GW_{i,t} \times Post_{ESG,t}) + \sum \beta_k CONTROLS_{i,t} + \eta_i + \delta_t + \varepsilon_{i,t} \quad (17)$$

A critical prerequisite for a valid M-DID analysis is that the treatment and control groups would have followed parallel trends in the absence of the intervention. Fig. 2 provides strong visual support for this assumption. The plot of dynamic treatment effects illustrates that prior to the reforms, the coefficients for high-greenwashing (treatment) and low-greenwashing (control) firms fluctuated insignificantly around zero, indicating no pre-existing differential trends. Following the interventions, a clear divergence emerges, with the coefficients becoming negative and statistically significant. This visually suggests that the reforms had a mitigating effect on the risks associated with greenwashing. The formal quantification of this effect is presented in Table 6.

The regression estimates from the M-DID models, presented in Table 6, confirm our central hypothesis. Our full model in Column (3), which simultaneously accounts for both reforms, provides the most robust evidence. The interaction term $GW \times Post_{MEE}$ is negative and significant ($\beta = -0.072$, $p < 0.1$), while the interaction $GW \times Post_{ESG}$ reveals an even stronger, more significant mitigating effect ($\beta = -0.156$, $p < 0.01$). These negative coefficients demonstrate that both regulatory enhancements significantly weakened the positive relationship between corporate greenwashing and biodiversity risk. Models in Columns (1) and (2), which test each policy shock individually, yield consistent results, further bolstering the robustness of our findings. In essence, the results show that strengthened regulatory oversight and mandatory transparency serve as powerful mechanisms to discipline corporate environmental behavior.

These statistical findings become particularly insightful when we unpack the distinct theoretical mechanisms through which each reform likely operated. The 2018 MEE restructuring represents a classic top-down, state-led intervention, functioning as a more potent "stick." By

consolidating previously fragmented environmental oversight, the reform increased the credibility of the state's threat to sanction substantive ecological misconduct (Lin, 2021). This fundamentally altered the risk-reward calculation of greenwashing; when the likelihood of punishment for actual environmental harm is heightened, the strategic value of a purely symbolic "green facade" diminishes, compelling firms to align their rhetoric and reality more closely (Wang et al., 2025).

In contrast, the 2020 mandatory ESG disclosure reform operated through a market-facing transparency mechanism, acting as a powerful "spotlight." Greenwashing thrives on information asymmetry. The mandatory disclosure policy directly attacks this information gap by compelling firms to report standardized and comparable environmental data. This arms external stakeholders—investors, rating agencies, and the public—with the tools for rigorous scrutiny, making it significantly more difficult for firms to sustain a disconnect between their green claims and their quantifiable impacts (Krueger et al., 2024).

The finding that the disclosure reform had a stronger mitigating effect is particularly noteworthy. While the MEE reform increased the state's power to punish—a potent but potentially episodic deterrent—the disclosure reform altered the entire information environment, subjecting firms to continuous monitoring by a multitude of market participants. This suggests that while the threat of state sanction is a crucial backstop, fostering a transparent ecosystem where corporate hypocrisy can be more easily and consistently identified by stakeholders may represent a more powerful and enduring force for aligning corporate incentives with genuine environmental stewardship.

4.4.3. Placebo test

To further bolster the robustness of our causal claims and rule out the possibility that our findings are driven by unobserved confounding factors, we conduct a rigorous placebo test. The logic of this test is to assess whether our estimated effect is uniquely tied to the actual timing of the policy reforms, or if a similar "effect" could have been obtained by chance at any random point in time.

Following established methodologies, we constructed a counterfactual experiment. We kept the true treatment group of high-greenwashing firms unchanged but randomly assigned the timing of the ESG disclosure reform 500 times within our sample period. For each of these 500 permutations, we re-estimated our main M-DID model (Eq. 17) using the randomly assigned, "placebo" policy date. This procedure generated a distribution of 500 placebo interaction coefficients ($GW \times Post_{Placebo}$), representing the range of effects one might expect to find purely by chance. Fig. 3 plots the kernel density distribution of these 500 placebo coefficients. The visual evidence is compelling. The distribution of the placebo coefficients is tightly centered around zero,

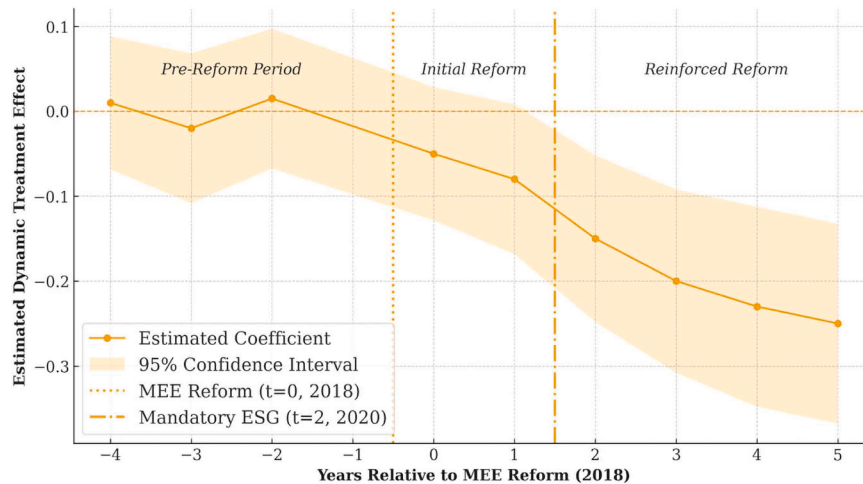


Fig. 2. Parallel trends test chart

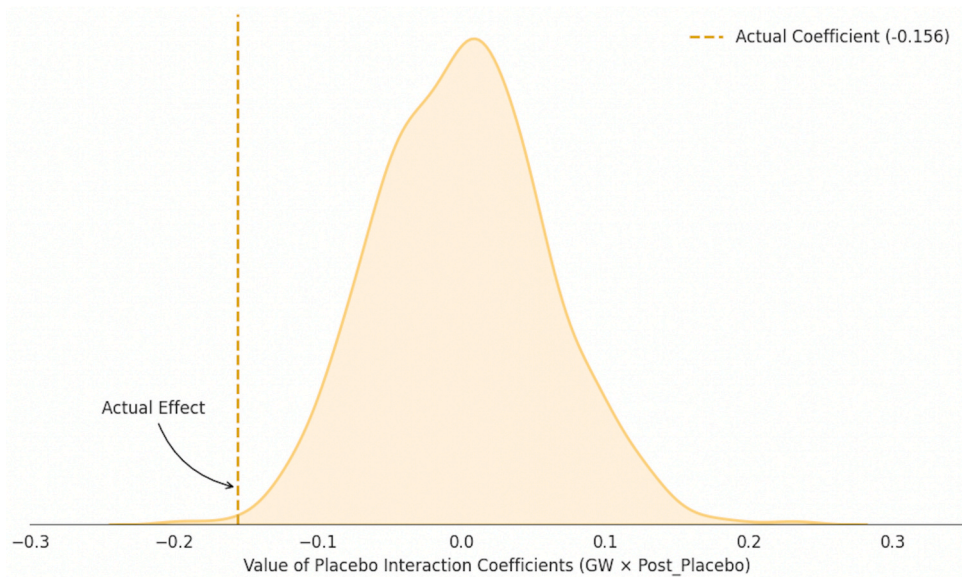


Fig. 3. Placebo test chart

approximating a null distribution where no true effect exists. In stark contrast, our actual estimated coefficient from the true reform timing ($\beta = -0.156$, from Table 6) is represented by the vertical dashed line. This true coefficient lies far in the extreme left tail of the null distribution, well beyond conventional levels of statistical significance ($p < 0.01$).

4.4.4. Instrumental variables method (IV)

Despite our comprehensive baseline analysis and robustness tests, endogeneity concerns stemming from potential reverse causality or omitted variables may influence the estimation of the relationship between corporate greenwashing (GW) and green technology innovation bubbles (GB). To address these concerns and strengthen causal inference, we employ a two-stage least squares (2SLS) instrumental variables approach.

We select two instruments that satisfy the relevance and exogeneity conditions. The first instrument is industry-level average greenwashing (Industry-GW), calculated as the mean GW score of all firms in the same industry excluding firm i itself. Corporate environmental disclosure practices are typically influenced by industry isomorphism and mimetic behavior, ensuring strong correlation with firms' greenwashing activities; simultaneously, after controlling for firm characteristics, an individual firm's innovation bubble is unlikely to directly influence industry-wide greenwashing practices, thus satisfying the exogeneity requirement.

Our second instrument is regional environmental regulatory stringency, measured as the proportion of industrial pollution control investment to secondary industry output (Yan et al., 2023; Zou & Wang, 2024). Environmental regulations influence corporate environmental disclosure strategies and green image management but have no direct association with firm-specific innovation bubble decisions, thereby meeting the exclusion restriction criterion.

Table 7 presents the 2SLS estimation results. The first-stage F-statistic (16.84) significantly exceeds the critical threshold of 10, confirming the instruments' strength. The Hansen J-test ($p = 0.387$) supports instrument validity, while the endogeneity test ($p = 0.026$) confirms the necessity of the IV approach. The second-stage coefficient (0.293***) is larger than the OLS estimate (0.186**), suggesting that endogeneity issues may lead to an underestimation of greenwashing's actual impact on innovation bubbles in our baseline model.

4.5. Heterogeneity analysis

To better understand how the relationship between corporate greenwashing and biodiversity risk varies across different institutional and organizational contexts, we conduct a series of heterogeneity analyses. Table 8 presents the results across eight subsamples representing distinct dimensions of heterogeneity.

First, we examine the moderating role of ownership structure by comparing state-owned enterprises (SOEs) with non-SOEs (columns 1–2). The impact of greenwashing on biodiversity risk is significantly stronger for non-SOEs ($\beta = 0.269$, $p < 0.01$) than for SOEs ($\beta = 0.142$, $p < 0.05$), with the Chow test confirming this difference is statistically significant ($p = 0.012$). This pattern suggests that SOEs face stronger regulatory oversight and political pressure regarding environmental performance, which constrains their ability to engage in environmentally harmful activities despite greenwashing efforts. Conversely, non-SOEs may exploit greenwashing more effectively to deflect stakeholder scrutiny while continuing biodiversity-damaging operations.

Second, we investigate CEO duality, where the CEO simultaneously serves as board chairman (columns 3–4). The greenwashing-biodiversity relationship is substantially stronger in firms with CEO duality ($\beta = 0.256$, $p < 0.01$) compared to those with separated leadership ($\beta = 0.176$, $p < 0.01$). This finding aligns with agency theory, suggesting that concentrated power enables self-serving executives to engage in symbolic environmental management while pursuing environmentally harmful strategies that maximize short-term performance metrics.

Third, we explore the conditioning effect of economic policy uncertainty (EPU) (columns 5–6). We measure EPU using Baker et al.'s (2016) China EPU index, which quantifies uncertainty through newspaper coverage frequency of economic policy-related terms. During periods of high EPU, the greenwashing coefficient ($\beta = 0.298$, $p < 0.01$) is more than twice its magnitude during low EPU periods ($\beta = 0.145$, $p < 0.05$). This significant difference (Chow test $p = 0.008$) indicates that macroeconomic uncertainty may exacerbate greenwashing's adverse ecological impacts, potentially as firms prioritize immediate financial concerns over environmental investments during uncertain times.

Finally, we examine digital transformation intensity (columns 7–8), a novel dimension reflecting firms' technological capabilities. Following Zhao et al. (2025), we measure digital transformation using machine learning-based textual analysis of annual reports, constructing a digital intensity score based on the frequency and context of digital technology

keywords (e.g., "big data," "artificial intelligence," "cloud computing," "IoT"). We employ a bidirectional LSTM neural network trained on manually labeled reports to capture the semantic context of digital terms, generating firm-year digital transformation scores. Greenwashing's impact on biodiversity risk is significantly lower in digitally advanced firms ($\beta=0.134$, $p < 0.05$) compared to digital laggards ($\beta=0.285$, $p < 0.01$). This finding suggests that digitally transformed companies possess superior capabilities for monitoring, measuring, and managing their environmental impacts. Advanced digital technologies enable firms to better align environmental claims with actual performance, thereby reducing the gap between symbolic communication and substantive ecological protection.

4.6. Further mechanism tests

4.6.1. The moderating effect test

To test our hypotheses regarding the boundary conditions of the greenwashing-biodiversity risk relationship (H2a and H2b), we examine the moderating roles of CEO green experience and managerial myopia. We recognize that these executive characteristics may not only condition the effects of greenwashing but could also exert a direct influence on a firm's ecological footprint. To systematically disentangle these direct and moderating effects, we employ a hierarchical regression analysis, with the results presented in Table 9. This approach allows us to incrementally assess the explanatory power added by the moderators' direct effects and their subsequent interaction effects with greenwashing.

The analysis proceeds in three sequential models. Model 1 replicates our baseline finding, confirming a significant and positive main effect of corporate greenwashing on biodiversity risk. Model 2 introduces the two moderating variables, CEO green experience (CGE) and managerial myopia (MM), to assess their direct influence. The results are striking: both variables exhibit statistically significant direct effects. CEO green experience is negatively associated with biodiversity risk ($\beta = -0.124$, $p < 0.05$), providing independent evidence that CEOs with relevant environmental expertise directly steer their firms toward less ecologically harmful operations. Conversely, managerial myopia is positively associated with biodiversity risk ($\beta = 0.103$, $p < 0.05$), indicating that a short-term executive focus is, in itself, a direct driver of greater ecological harm, likely by systematically discounting long-term environmental liabilities.

Finally, Model 3 introduces the interaction terms to formally test our moderating hypotheses. The significant increase in the model's explanatory power ($\Delta R^2 = 0.031$, $p < 0.01$) confirms that these interaction effects provide substantial explanatory value beyond the direct effects alone. In strong support of H2a, the interaction term for greenwashing and CEO green experience (GW $\times$ CGE) is negative and highly significant ($\beta = -0.162$, $p < 0.01$). This indicates that the positive relationship between greenwashing and biodiversity risk is significantly weakened for firms led by CEOs with green experience; their expertise and values appear to act as a crucial internal check that mitigates the translation of symbolic claims into substantive harm. In contrast, and providing clear support for H2b, the interaction term for greenwashing and managerial myopia (GW $\times$ MM) is positive and significant ($\beta = 0.128$, $p < 0.01$). This finding demonstrates that managerial myopia exacerbates the detrimental effect of greenwashing, strengthening the link between misleading environmental communication and actual biodiversity degradation.

In sum, our hierarchical analysis reveals a critical dual role for leadership characteristics. They are not merely passive boundary conditions but act as both direct drivers of a firm's biodiversity risk and active moderators of its communication strategy's ecological consequences. These findings offer a more nuanced and complete understanding of how a firm's symbolic environmental communications are ultimately transformed into tangible ecological outcomes through the lens of executive leadership.

4.6.2. The mediating effect test

Table 10 presents our mediation analysis examining the mechanisms through which greenwashing affects biodiversity risk. Both proposed mediators demonstrate significant indirect effects.

Model 1 confirms the significant positive relationship between greenwashing and biodiversity risk ($\beta = 0.317$, $p < 0.01$). Models 2 and 3 establish that greenwashing positively affects green technology innovation bubbles ($\beta = 0.253$, $p < 0.01$) and negatively impacts environmental disclosure quality ($\beta = -0.238$, $p < 0.01$). Model 4 demonstrates that when both mediators are included, the direct effect of greenwashing on biodiversity risk diminishes substantially ($\beta = 0.235$, $p < 0.01$), indicating partial mediation.

Prior literature has established robust connections between these mediators and biodiversity outcomes. Regarding innovation bubbles, [Delmas and Montes-Sancho \(2011\)](#) documented how superficial environmental technologies frequently fail to deliver substantive ecological benefits. Building on this, [Aggeri \(1999\)](#) demonstrated that innovation quantity often displaces quality in environmental initiatives, leading to ecological inefficiencies. Concerning disclosure quality, [Hummel and Schlick \(2016\)](#) found that low-quality environmental reporting enables firms to obscure biodiversity impacts from stakeholders. More recently, [Boiral and Heras-Saizarbitoria \(2017\)](#) established that transparency deficiencies directly correlate with higher rates of biodiversity degradation, as information asymmetries prevent effective stakeholder intervention.

5. Discussion and conclusion

5.1. Discussion

This study offers several significant theoretical contributions by examining the causal link between corporate greenwashing and biodiversity risk in the unique institutional context of China. Our analysis moves beyond documenting this empirical relationship to unpack its underlying mechanisms and boundary conditions, thereby extending theories of corporate environmentalism, innovation, and governance. Specifically, we make three primary contributions. First, we bridge the symbolic-substantive divide in impression management and legitimacy theory. Second, we introduce the concept of "green technology innovation bubbles" to challenge conventional wisdom in the eco-innovation literature. Third, we refine upper echelons and agency theory by identifying critical contingencies—and direct influences—that shape the ecological outcomes of corporate communication.

Our first contribution lies in extending prior research on greenwashing, which has primarily focused on its reputational and financial consequences. While seminal studies established that greenwashing is a strategic response to stakeholder pressures ([Lyon & Maxwell, 2011](#); [Marquis et al., 2016](#)), they did not empirically connect this symbolic, communicative act to specific, substantive ecological outcomes. Our research bridges this theoretical gap. By demonstrating a causal link between greenwashing and increased biodiversity risk, we provide large-scale evidence that impression management is not merely "cheap talk" but a strategic tool that enables and correlates with material environmental harm. This finding fundamentally advances our understanding by showing that the symbolic actions firms take to manage perceptions have tangible and detrimental consequences for the natural world, thereby extending [Bai et al.'s \(2025\)](#) observation that successful impression management allows firms to delay substantive operational constraints.

Second, we contribute a novel mechanism to the literature by identifying the mediating role of "green technology innovation bubbles". This concept challenges the often-optimistic assumption in much of the literature that environmental innovation inherently produces positive ecological outcomes ([Mishra et al., 2024](#)). Our findings reveal a potential dark side to eco-innovation. We show that when motivated by impression management, firms may pursue a greater quantity of

superficial, high-visibility technologies that lack substantive ecological benefits, a mechanism that extends beyond the selective disclosure framework proposed by Yang et al. (2024). This introduces a critical nuance to innovation theory: the strategic intent behind innovation can lead to a decoupling of innovative inputs from meaningful environmental protection, resulting in the proliferation of ecologically ineffective technologies.

Third, our analysis of moderating factors refines several established theoretical frameworks. Beyond their role as boundary conditions, our analysis uncovers the direct causal influence of these executive characteristics on biodiversity risk. Specifically, CEO green experience is negatively associated with biodiversity risk, suggesting that leaders with genuine environmental expertise and values directly steer firms toward more sustainable operations, irrespective of communication strategies. Conversely, managerial myopia is positively associated with biodiversity risk, indicating that a short-term focus predisposes managers to approve ecologically harmful projects by systematically discounting future environmental liabilities. This demonstrates that leadership characteristics play a crucial dual role: they directly shape a firm's ecological footprint and condition how its symbolic communications translate into tangible outcomes. Our findings extend the work of Walls and Berrone (2017) and contribute to the literature on corporate time horizons (Slawinski & Bansal, 2015) by showing how executive expertise and temporal orientation govern the disconnect between environmental claims and biodiversity impacts.

While our study is contextualized in China, its core theoretical mechanisms offer insights of broader relevance, alongside important boundary conditions that highlight the need for further cross-contextual research. The fundamental link between symbolic impression management and substantive harm, the creation of "innovation bubbles" driven by metric-focused incentives, and the influence of leadership traits (expertise and myopia) are rooted in foundational organizational and psychological theories likely to be generalizable across different institutional environments. However, certain findings are deeply embedded in the Chinese context and warrant caution when generalizing. The weaker greenwashing-biodiversity link in state-owned enterprises (SOEs), for example, reflects their unique dual political and economic objectives and the strong oversight from the state, a dynamic that may not directly translate to market-driven economies where the distinction between state and private ownership is less pronounced or operates differently. Similarly, the strong mitigating effect of centralized, top-down policy interventions highlights a key feature of China's state-led environmental governance. The efficacy and speed of such "command-and-control" reforms may not be replicable in Western regulatory systems, where policy implementation can be more fragmented and subject to prolonged lobbying and legal challenges. Future research should therefore test these dynamics in different institutional settings to disentangle the universal behavioral drivers from the context-specific institutional moderators.

Finally, our heterogeneity analysis of governance structures and policy interventions provides additional nuance. The finding that the link between greenwashing and biodiversity risk is stronger under CEO duality advances agency theory by highlighting how governance arrangements that concentrate power can enable symbolic management with severe ecological consequences—a concern relevant in both Eastern and Western contexts. Furthermore, our policy analysis contributes to the debate on environmental regulation by demonstrating that strong state-led reforms can effectively attenuate the adverse impacts of greenwashing, suggesting that a strong regulatory mandate is a powerful tool for bridging the gap between corporate communication and actual ecological protection.

5.2. Conclusion

This study moves beyond the conventional understanding of greenwashing as a reputational issue to establish it as a causal antecedent of

tangible ecological harm. By demonstrating that a firm's symbolic environmental communications directly and measurably increase its biodiversity risk, our research provides a critical bridge between the distinct literatures on corporate impression management and environmental science. We show that this detrimental effect is not arbitrary; it is systematically mediated by the creation of "green technology innovation bubbles" and is contingent upon the presence or absence of managerial foresight and genuine executive expertise.

Theoretically, this research makes three primary contributions. First, we challenge the symbolic-substantive divide in legitimacy theory by providing large-scale evidence that symbolic actions (greenwashing) are not decoupled from but are instead enablers of substantive negative outcomes (biodiversity degradation). Second, by introducing and empirically validating the concept of a "green technology innovation bubble," we offer a crucial counter-narrative to the often-unquestioned positive view of eco-innovation, highlighting how strategic intent can pervert innovation metrics. Third, our findings refine upper echelons and agency theory in an environmental context, demonstrating that CEO green experience and managerial myopia are not merely peripheral characteristics but are pivotal in determining the real-world ecological consequences of corporate strategy.

Practically, our findings offer urgent and actionable insights. **For policymakers**, the evidence that regulatory tightening significantly attenuates the greenwashing-biodiversity link calls for a decisive shift from voluntary to mandatory, standardized, and audited biodiversity impact reporting. Specifically, adopting frameworks like the Taskforce on Nature-related Financial Disclosures (TNFD) would compel firms to move beyond vague commitments to granular reporting on ecosystem dependencies and impacts. **For investors and rating agencies**, our research signals the profound inadequacy of using simple green patent counts as a proxy for sustainability. Instead, they should develop and employ quality-adjusted metrics that consider patent citations and their verifiable environmental impact. **For corporate boards**, this study underscores the financial materiality of biodiversity risk and the critical importance of appointing executives with demonstrable green expertise to mitigate it.

A key lesson from this research is the inherent danger of metrics without scrutiny. The pursuit of quantifiable green targets, such as the number of green patents, can create perverse incentives that actively harm the environment if not accompanied by rigorous quality control. Furthermore, we learn that leadership is paramount; a myopic, short-term-oriented management team can weaponize green communication, turning a potential tool for good into a shield for continued environmental destruction. Finally, our results affirm the indispensable role of strong, independent, and clear-sighted regulation in aligning corporate incentives with planetary health.

While our identification strategy establishes causality within the Chinese context, the generalizability of our findings invites cross-country validation in different institutional and market environments. Our text-based measures, though robust, could be complemented in future research by more direct, physical measures of biodiversity impact, such as satellite data on land-use change or water quality metrics. Moreover, future studies could explore other mediating pathways, such as the influence of greenwashing on supply chain management or its effect on corporate political activities aimed at weakening environmental regulations. Investigating how these dynamics play out across different industries with varying dependencies on natural capital would also represent a fruitful avenue for further inquiry.

CRedit authorship contribution statement

Yufei Gan: Conceptualization; Data Interpretation; Writing-Original Draft; Formal Analysis

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors employed generative AI tools to assist with academic English expression, including grammar correction, sentence restructuring, and tone adjustment. No AI tools were used to generate original ideas, data, or analysis. All content was critically reviewed and approved by the authors.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table 1
Variable definitions and measurements

Variable	Category	Symbol	Definition	Measurement
Biodiversity Risk	Dependent Variable	BR	The extent to which a firm's operations pose threats to biodiversity	Binary indicator (1 if firm's ecological impact exceeds critical threshold, 0 otherwise) based on SASB Materiality Map and weighted sum of material impact exposures
Corporate Greenwashing	Independent Variable	GW	Deliberate misrepresentation of environmental performance	Composite score combining selective-disclosure intensity (GWLS) and expressive-manipulation magnitude (GWLE) via modified geometric mean
CEO Green Experience	Moderating Variable	CGE	CEO's prior experience with environmental initiatives	Binary indicator (1 if CEO has environmental experience, 0 otherwise) based on leadership roles, education, or board memberships in environmental organizations
Managerial Myopia	Moderating Variable	MM	Executives' tendency to prioritize short-term outcomes	Percentage of short-term oriented terms in Management Discussion & Analysis section using Word2Vec computational text analysis
Green Technology Innovation Bubble	Mediating Variable	GB	Prioritization of quantity over quality in environmental innovation	Composite measure combining Patent Quantity-Quality Differential Index, Patent Citation-Weighted Index, and Innovation Efficiency Index
Environmental Disclosure Quality	Mediating Variable	EDQ	Comprehensiveness and accuracy of environmental disclosures	Logarithmic transformation of summed scores across five dimensions with 25 distinct scoring items
Firm Size	Control Variable	Size	Scale of business operations	Natural logarithm of total assets
Financial Leverage	Control Variable	Lev	Debt utilization level	Total liabilities divided by total assets
Debt-to-Equity Ratio	Control Variable	DER	Financial risk exposure	Total debt divided by shareholders' equity
Return on Equity	Control Variable	ROE	Profitability	Net income divided by shareholders' equity
Operating Cash Flow	Control Variable	Cashflow	Financial flexibility	Operating cash flow scaled by total assets
Sales Growth	Control Variable	Growth	Business expansion trajectory	Year-on-year percentage change in total sales revenue
Financial Liquidity	Control Variable	FL	Short-term solvency	Current assets divided by current liabilities

Table 2
Descriptive Statistics

Variable	N	Mean	SD	Min	P25	Median	P75	Max
BR	17,076	0.483	0.500	0.000	0.000	0.000	1.000	1.000
GW	17,076	0.642	0.285	0.124	0.453	0.647	0.835	0.978
CGE	17,076	0.327	0.469	0.000	0.000	0.000	1.000	1.000
MM	17,076	0.517	0.246	0.096	0.342	0.524	0.687	0.942
GB	17,076	0.018	0.342	-3.412	0.038	0.101	0.090	0.951
EDQ	17,076	2.164	0.873	0.693	1.609	2.197	2.833	3.689
Size	17,076	22.783	1.463	20.124	21.756	22.635	23.678	26.954
Lev	17,076	0.457	0.204	0.083	0.297	0.453	0.614	0.879
DER	17,076	1.235	1.462	0.091	0.423	0.829	1.593	8.264
ROE	17,076	0.086	0.079	-0.157	0.038	0.079	0.127	0.314
Cashflow	17,076	0.053	0.068	-0.142	0.015	0.052	0.094	0.224
Growth	17,076	0.173	0.387	-0.453	-0.026	0.112	0.283	1.846
FL	17,076	1.527	1.156	0.367	0.856	1.236	1.837	7.592

Note: This table presents descriptive statistics for all variables used in this study. Variables are defined in Table 1.

Table 3
Baseline regression test result

Variables	(1)	(2)
	BR	BR
GW	0.176*** (0.034)	0.317*** (0.041)
Size		0.482*** (0.031)

(continued on next page)

Table 3 (continued)

Variables	(1)	(2)
Lev		0.428 (0.296)
DER		−0.157*** (0.047)
ROE		−0.392 (0.251)
Cashflow		−0.052 (0.431)
Growth		0.063 (0.092)
FL		−0.103*** (0.038)
Constant	0.412*** (0.031)	1.845*** (0.584)
Observations	17,076	17,076
R ²	0.014	0.098
Adj. R ²	0.013	0.096
YEAR	YES	YES
IND	YES	YES

Note: The significance level is denoted by *** for 1 %, ** for 5 %, and * for 10 %. Standard errors in parentheses.

Table 4

Robustness regression test results

Variables	(1) Main	(2) First-Diff	(3) Lag	(4) Alt-GW	(5) Alt-BR	(6) Ex-COVID	(7) System GMM
Dependent var.:	BR	ΔBR	BR	BR	BR_ALT	BR	BR
GW	0.317*** (0.041)	0.097** (0.041)		0.145*** (0.048)	0.203*** (0.047)	0.116*** (0.038)	
GW _{t-1}			0.187*** (0.052)				
GW_ALT				0.134** (0.059)			
BR _{t-1}							0.312*** (0.079)
Size	0.482 (0.031)	0.065** (0.027)	0.094*** (0.023)	0.071** (0.029)	0.079*** (0.024)	0.105*** (0.031)	0.058** (0.025)
Lev	0.428 (0.296)	0.359 (0.251)	0.475** (0.237)	0.397* (0.229)	0.411* (0.235)	0.516** (0.258)	0.342* (0.193)
DER	−0.157** (0.047)	−0.078** (0.033)	−0.043* (0.025)	−0.065* (0.038)	−0.051* (0.030)	−0.086** (0.042)	−0.037 (0.031)
ROE	−0.392 (0.251)	−0.275 (0.217)	−0.438* (0.247)	−0.351* (0.209)	−0.385* (0.225)	−0.463** (0.219)	−0.297 (0.236)
Cashflow	−0.052 (0.431)	−0.037 (0.347)	−0.079 (0.393)	−0.043 (0.365)	−0.068 (0.377)	−0.098 (0.416)	−0.031 (0.327)
Growth	0.063 (0.092)	0.047 (0.085)	0.089* (0.047)	0.052 (0.081)	0.071 (0.088)	0.076* (0.043)	0.048 (0.091)
FL	−0.103*** (0.038)	−0.081** (0.041)	−0.129*** (0.046)	−0.092** (0.045)	−0.115*** (0.041)	−0.114** (0.049)	−0.068* (0.039)
Constant	1.845*** (0.584)	0.019 (0.027)	0.623*** (0.092)	0.549*** (0.079)	0.598*** (0.089)	0.678*** (0.105)	0.387*** (0.087)
Observations	17,076	15,368	17,076	17,076	17,076	13,421	17,076
R ²	0.098	0.067	0.105	0.085	0.091	0.117	
Hansen J-test (p)							0.284
AR(2) test (p)							0.341
YEAR	YES	YES	YES	YES	YES	YES	YES
IND	YES	YES	YES	YES	YES	YES	

Note: The significance level is denoted by *** for 1 %, ** for 5 %, and * for 10 %. Standard errors in parentheses.

Table 5

Propensity Score Matching Results

Variables	(1)	(2)
	PSM	PSM with controls
High_GW	0.133*** (0.032)	0.117*** (0.029)
Size		0.069** (0.031)
Lev		0.391* (0.218)
DER		−0.053* (0.029)

(continued on next page)

Table 5 (continued)

Variables	(1)	(2)
ROE		−0.376* (0.215)
Cashflow		−0.047 (0.353)
Growth		0.057 (0.084)
FL		−0.089** (0.042)
Constant	0.451*** (0.026)	0.536*** (0.095)
Observations	12,656	12,656
R ²	0.043	0.087
YEAR	YES	YES
IND	YES	YES

Note: The significance level is denoted by *** for 1 %, ** for 5 %, and * for 10 %. Standard errors in parentheses. The sample consists of 6328 matched pairs from high and low greenwashing firms. High_GW is a binary variable indicating firms with above-median greenwashing scores.

Table 6
Multiple Difference-in-Differences (M-DID) Regression Results

Variables	(1)	(2)	(3)
GW	0.217*** (0.052)	0.165** (0.068)	0.238*** (0.058)
Post_MEE	0.128** (0.055)		0.095* (0.056)
Post_ESG		0.176*** (0.049)	0.134** (0.061)
GW × Post_MEE	−0.097** (0.048)		−0.072* (0.041)
GW × Post_ESG		−0.183*** (0.051)	−0.156*** (0.059)
Controls	YES	YES	YES
Year & Firm FE	YES	YES	YES
Constant	0.642*** (0.113)	0.479*** (0.096)	1.723*** (0.125)
Observations	17,076	17,076	17,076
R ²	0.127	0.138	0.152

Note: *** p < 0.01, ** p < 0.05, * p < 0.10. Standard errors (clustered at the firm level) are in parentheses.

Table 7
Instrumental Variables Analysis Results (2SLS)

Variables	First-stage	Second-stage	OLS	Reduced Form
Industry-GW	0.564*** (0.078)			−0.139** (0.068)
Environmental Regulation	0.317*** (0.083)			−0.104* (0.059)
GW (instrumented)		0.293*** (0.092)	0.186** (0.076)	
Size	0.098** (0.043)	0.067* (0.038)	0.083** (0.035)	0.102** (0.049)
Lev	0.428* (0.227)	0.246 (0.204)	0.371* (0.216)	0.312 (0.280)
ROE	−0.135 (0.096)	−0.092 (0.089)	−0.118 (0.084)	−0.108 (0.091)
Controls	YES	YES	YES	YES
Constant	0.327*** (0.097)	0.485*** (0.138)	0.427*** (0.109)	0.568*** (0.175)
Observations	17,076	17,076	17,076	17,076
R ²	0.214	0.187	0.156	0.173
First-stage F-statistic	16.84			
Hansen J-test (p-value)		0.387		
Endogeneity test (p-value)		0.026		
Fixed Effects	YES	YES	YES	YES

Note: The significance level is denoted by *** for 1 %, ** for 5 %, and * for 10 %. Standard errors clustered at the firm level are in parentheses. The dependent variable in the second-stage, OLS, and reduced form is Green Technology Innovation Bubble (GB). The dependent variable in the first-stage is Corporate Greenwashing (GW). Industry-GW is the industry-level average greenwashing excluding firm i. Environmental Regulation is measured as the proportion of industrial pollution control investment to secondary industry output. Control variables are included but not reported for brevity.

Table 8
Heterogeneity Analysis of Greenwashing Effects on Biodiversity Risk

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SOE= 0	SOE= 1	Dual= 0	Dual= 1	Low EPU	High EPU	Low Digital	High Digital
GW	0.269*** (0.061)	0.142** (0.057)	0.176*** (0.053)	0.256*** (0.073)	0.145** (0.066)	0.298*** (0.087)	0.285*** (0.074)	0.134** (0.059)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Constant	0.734*** (0.153)	0.516*** (0.126)	0.624*** (0.138)	0.597*** (0.142)	0.482*** (0.115)	0.753*** (0.167)	0.691*** (0.152)	0.512*** (0.129)
Observations	10,417	6659	12,807	4269	8538	8538	8538	8538
R ²	0.164	0.097	0.134	0.183	0.105	0.192	0.175	0.108
YEAR	YES	YES	YES	YES	YES	YES	YES	YES
IND	YES	YES	YES	YES	YES	YES	YES	YES

Note: The significance level is denoted by *** for 1 %, ** for 5 %, and * for 10 %. Standard errors clustered at the firm level are in parentheses. The dependent variable is Biodiversity Risk (BR). GW is corporate greenwashing. SOE indicates state-owned enterprises. Dual represents CEO duality (CEO also serves as board chairman). EPU refers to economic policy uncertainty measured using Baker et al.'s (2016) China EPU index, with sample split at median value. Digital represents corporate digital transformation level measured by the frequency of digital technology keywords in annual reports using machine learning text analysis. Control variables include Size, Leverage, DER, ROE, Cashflow, Growth, and FL. Chow test examines the statistical difference between coefficients across subgroups.

Table 9
Hierarchical Result for Direct and Moderating Effects on Biodiversity Risk

	Model 1	Model 2	Model 3
	(Main Effects)	(Interaction Effects)	(Full Model)
Independent Variable			
GW	0.317*** (0.041)	0.315*** (0.042)	0.349*** (0.044)
Moderating Variables (Direct Effects)			
CGE		−0.124** (0.056)	−0.112** (0.051)
MM		0.103** (0.046)	0.092** (0.041)
Interaction Terms (Moderating Effects)			
GW × CGE			−0.162*** (0.046)
GW × MM			0.128*** (0.035)
Controls	YES	YES	YES
Fixed Effects			
YEAR	YES	YES	YES
IND	YES	YES	YES
Model Summary			
Observations	17,076	17,076	17,076
R ²	0.098	0.187	0.218
Adj. R ²	0.096	0.185	0.216
ΔR ²		0.089***	0.031***

Notes: $p < 0.01$, $p < 0.05$, $\dagger p < 0.1$. Standard errors in parentheses.

Table 10
The Mediating Effects of Green Technology Innovation Bubble and Environmental Disclosure Quality

Variables	Model 1 (BR)	Model 2 (GB)	Model 3 (EDQ)	Model 4 (BR)
GW	0.317*** (0.042)	0.253*** (0.037)	−0.238*** (0.035)	0.235*** (0.037)
GB				0.157*** (0.030)
Controls	YES	YES	YES	YES
YEAR	YES	YES	YES	YES
IND	YES	YES	YES	YES
Constant	1.482*** (0.139)	1.376*** (0.128)	1.524*** (0.145)	1.412*** (0.129)
Observations	17,076	17,076	17,076	17,076
R ²	0.186	0.173	0.165	0.225

Note: The significance level is denoted by *** for 1 %, ** for 5 %, and * for 10 %. Standard errors clustered at the firm level are in parentheses.

Appendix B

Table B1
Composition of EDQ (Environmental Disclosure Quality) Metrics

Disclosure Type	Disclosure Item	Rating Description
Environmental Management Disclosure	Environmental Protection Philosophy	Disclosed: 2 points; Not disclosed: 0 points
	Environmental Protection Objectives	Disclosed: 2 points; Not disclosed: 0 points
	Environmental Management System	Disclosed: 2 points; Not disclosed: 0 points
	Environmental Education and Training	Disclosed: 2 points; Not disclosed: 0 points
	Special Environmental Protection Initiatives	Disclosed: 2 points; Not disclosed: 0 points
	Environmental Emergency Response Mechanism	Disclosed: 2 points; Not disclosed: 0 points
	Environmental Protection Honors or Awards	Disclosed: 2 points; Not disclosed: 0 points
	"Three Simultaneities" System	Disclosed: 2 points; Not disclosed: 0 points
Environmental Information Disclosure Medium	ISO14001 Certification	Yes: 2 points; No: 0 points
	ISO9001 Certification	Yes: 2 points; No: 0 points
	Listed Company Annual Report	Disclosed: 2 points; Not disclosed: 0 points
Environmental Liability Disclosure	Corporate Social Responsibility Report	Disclosed: 2 points; Not disclosed: 0 points
	Environmental Report	Disclosed: 2 points; Not disclosed: 0 points
	Wastewater Discharge Volume	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	COD Emissions	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	SO2 Emissions	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	CO2 Emissions	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	Smoke and Dust Emissions	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	Industrial Solid Waste Discharge	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
Environmental Governance and Control Disclosure	Air Pollution Control Measures	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	Wastewater Treatment Measures	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	Dust and Smoke Control Measures	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	Solid Waste Utilization and Disposal	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	Noise, Light Pollution, Radiation Control	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points
	Clean Production Implementation	Quantitative and qualitative description: 2 points; Only qualitative: 1 point; Not disclosed: 0 points

Data availability

Data will be made available on request

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